# Traffic Crash Analytics & Safety Intelligence Platform

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Technologies: Python | SQLite | SQL | Pandas | Streamlit

# ABSTRACT

The Traffic Crash Analytics & Safety Intelligence Platform is a project built with the aid of Python, SQLite and Streamlit to analyze traffic accident data and provide useful information on road safety, trends in accidents, accident patterns, and accident causality risks. In this project, different advanced SQL methods have been used to find out risk hotspots, risky periods, and accident causality factors from the traffic data set. The findings and business intelligence obtained through the analysis are displayed through a Streamlit interface.

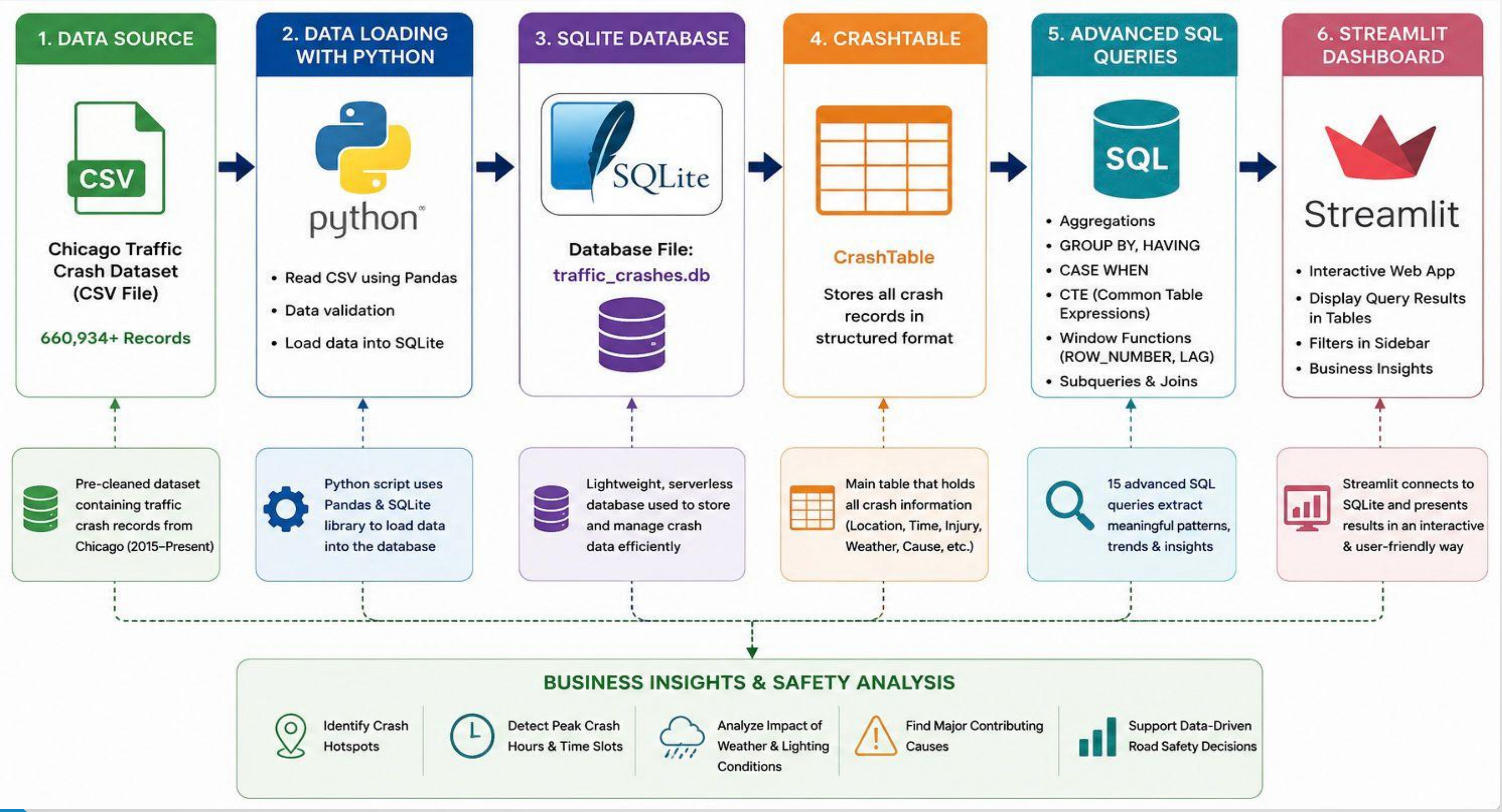
# PROBLEM STATEMENT

Traffic crash data contains valuable insights that can help improve road safety, optimize emergency response, and support policy decisions. However, extracting meaningful insights from large-scale structured data requires strong SQL and analytical skills.

# TECHNOLOGY STACK

- Python (version 3.12.4 ): Used to load the CSV data into SQLite, execute SQL queries. It was also used to connect the SQLite database with the Streamlit dashboard.
- SQLite (version3): Used to store and manage the traffic crash data in a structured relational database. It is lightweight, serverless, and easily integrates with Python, making it suitable for this analytics project
- Streamlit (version 1.58.0) : Used to create an interactive dashboard for displaying query results and business insights.

# ARCHITECTURE



# SQL ANALYSIS

**15 analytical SQL queries implemented**

**Advanced SQL Concepts Used**

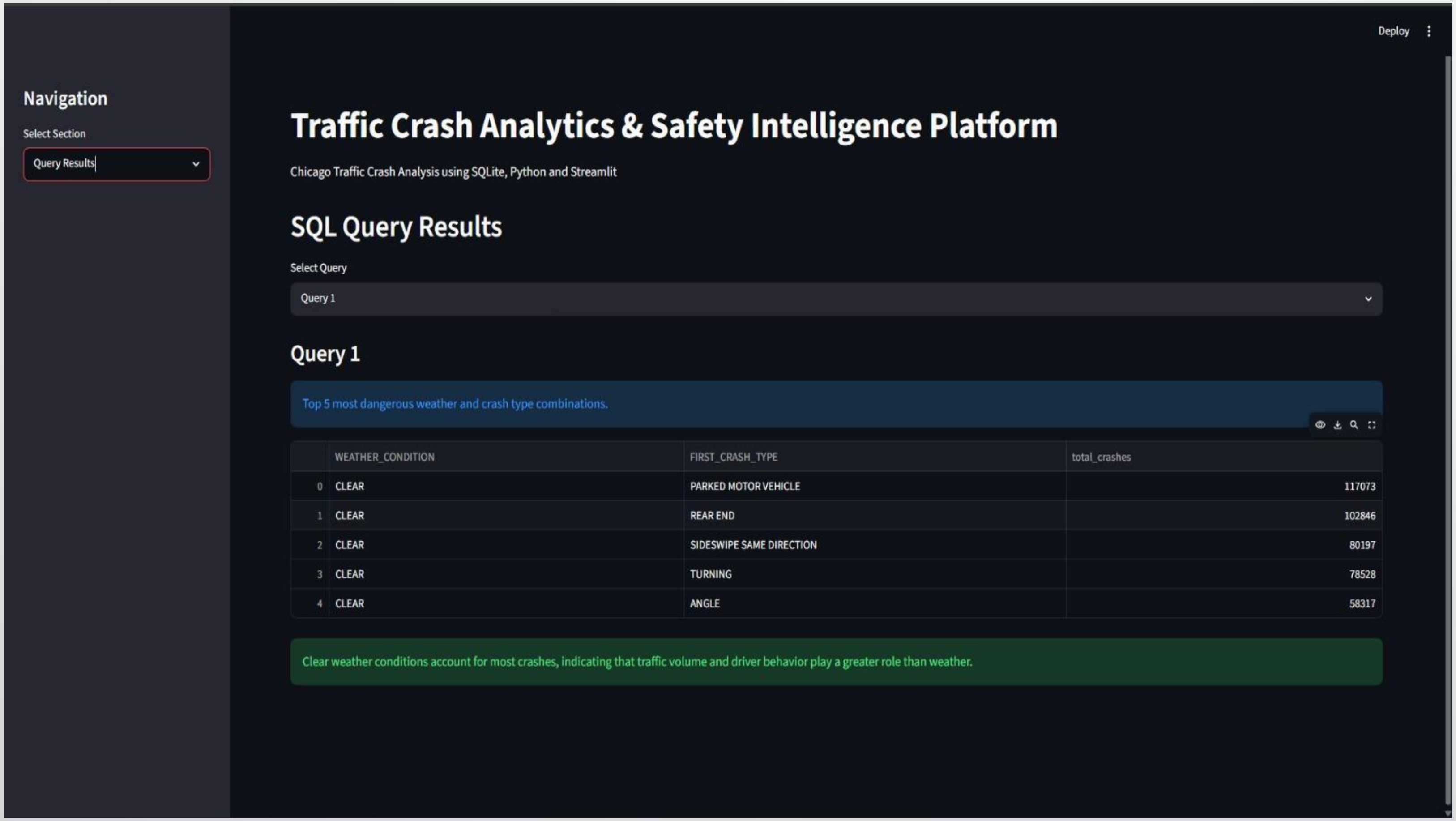
- GROUP BY()
- CASE()
- HAVING()
- CTE()
- ROW_NUMBER()
- LAG()

**Key Queries**

- Hotspot Analysis
- Year-over-Year Growth
- Top Contributing Causes
- Injury Rate Analysis

# STREAMLIT DASHBOARD

# SAMPLE QUERY RESULTS



Deploy

# Traffic Crash Analytics & Safety Intelligence Platform

Chicago Traffic Crash Analysis using SQLite, Python and Streamlit

**Navigation**

Select Section

Query Results

## SQL Query Results

Select Query

Query 14

### Query 14

Year-over-year crash growth analysis using LAG().

| | year | total_crashes | previous_year_crashes | growth_rate_percent |
|---|---|---|---|---|
| 0 | 2020 | 91509 | None | None |
| 1 | 2021 | 107937 | 91509 | 17.95 |
| 2 | 2022 | 107462 | 107937 | -0.44 |
| 3 | 2023 | 109714 | 107462 | 2.1 |
| 4 | 2024 | 110853 | 109714 | 1.04 |
| 5 | 2025 | 107965 | 110853 | -2.61 |
| 6 | 2026 | 25494 | 107965 | -76.39 |

Growth trends help assess whether road safety initiatives are improving or worsening crash outcomes.

# CONCLUSION

- Successfully analyzed 660,934 traffic crash records using Python, SQLite, and Streamlit.
- Identified crash hotspots, injury patterns, peak crash periods, and major contributing causes through advanced SQL analysis.
- Developed an interactive dashboard that converts raw crash data into meaningful insights for road safety planning.
- The project demonstrates how data analytics can support data-driven decisions to improve traffic safety and reduce accident risks.